# Rearrangement of a formula

1 Make $x$ the subject of the following equations:

a $y = x + z$

b $y = \dfrac{x}{-6}$

c $y = 7(2 + x)$

d $\dfrac{x}{2} + \dfrac{n}{5} = 2$

e $3y = 2x + z$

f $y = \dfrac{-x}{7}$

g $y = 5(3 - x)$

h $\dfrac{x}{5} - \dfrac{m}{7} = 2$

2 Make $m$ the subject of the following equations:

a $\dfrac{m}{y} = gh$

b $y = 4mx - 9$

c $x = 5k(n + m)$

d $\dfrac{2m}{p} = st$

e $b = -2am + 10$

f $x = -7m(p + q)$

g $y = -5mx - 14$

h $x = -7k(l - m)$

3 Make $y$ the subject of the following equations:

a $\dfrac{x}{y} = h - k$

b $9x^2 + 3y = 12$

c $\dfrac{3a}{b} = 2j - y$

d $4x^2 - 2y = -16$

4 Make $k$ the subject of the following equations:

a $m = \dfrac{9kx}{8} - 6$

b $r = \dfrac{k}{k - 9}$

c $v = \dfrac{5kw}{7} - 3$

d $s = \dfrac{2k}{k + 4}$

5 Make $R$ the subject of $V = IR - E$.

6 Make $r$ the subject of $L = \dfrac{E}{R + r}$.

7 Make $p$ the subject of the equation $M = pq - 5pr^2$.

8 Consider the Pythagorean formula $c^2 = a^2 + b^2$.

a Make $a^2$ the subject of the formula.

b Make $b^2$ the subject of the formula.

9 Make $r^2$ the subject of the following formulas:

a The volume of a cone is given by the formula $V = \dfrac{1}{3}\pi r^2 h$.

b The area of a circle is given by the formula $A = \pi r^2$.

**c** The area of a sector of a circle is given by the formula $A = \dfrac{\theta \pi r^2}{360}$.

**10** Make $v^2$ the subject of the formula for kinetic energy, $K = \dfrac{1}{2}mv^2$.

**11** The bend allowance for sheet metal is given by the formula $B = 2\pi \left(R + \dfrac{T}{2}\right) \times \dfrac{A}{360}$.

Make $T$, the thickness of the sheet, the subject of the formula.

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## Applications

**12** Temperature conversion from degrees Fahrenheit to degrees Celsius is given by the formula $C = \dfrac{5}{9}(F - 32)$.

The temperature inside a freezer is $5^\circ\text{F}$. Find this temperature in degrees Celsius.

**13** The perimeter of a rectangle is given by the formula $P = 2(a + b)$.

Find the value of $b$ if $a = 5$ and $P = 26$.

**14** The surface area of a rectangular prism is given by the formula $S = 2(ab + bh + ah)$.

Find the value of $h$ if $a = 3$, $b = 3$ and $S = 138$.

**15** The surface area of a rectangular prism is given by the formula $S = 2(lw + wh + lh)$, where $l$, $w$ and $h$ are the dimensions of the prism.

Find the surface area of a rectangular prism with a length of $8 \text{ cm}$, a width of $9 \text{ cm}$ and a height of $5 \text{ cm}$.

**16** The surface area of a cylinder is given by the formula $S = 2\pi r^2 + 2\pi rh$.

Find the surface area of a cylinder with a diameter of $10 \text{ cm}$ and a height of $25 \text{ cm}$.

**17** The volume of a sphere is given by the formula $V = \dfrac{4}{3}\pi r^3$, where $r$ is the radius of the sphere. Find the radius of a sphere with a volume of $124 \text{ mm}^3$ to two decimal places.

**18** Newton's second equation of motion is $s = ut + \dfrac{1}{2}at^2$.

Find the value of $a$ if $s = 2596$, $u = 250$ and $t = 22$.

**19** The power of a circuit is given by the formula $P = I^2 R$.

Find the value of $I$, correct to two decima[redacted]es, when $P = 40$ and $R = 560$.

**20** The resistance of a parallel circuit is given by the formula $\frac{1}{a} = \frac{1}{b} + \frac{1}{c}$.

Find the value of $b$, correct to two decimal places, if $a = 8$ and $c = 23$.

**21** The reactance of a capacitor is given by the formula: $X = \frac{1}{2\pi f C}$.

Find the value of $f$, correct to one decimal place, if $X = 44.1$ and $C = 190 \times 10^{-9}$.

**22** The gravitational force between two masses is given by $F = \frac{GMm}{d^2}$.

Find the value of $M$, correct to two decimal places, if $F = 15$, $G = 6.67 \times 10^{-11}$, $m = 1$ and $d = 5 \times 10^{-6}$.

**23** Amelia bought a television series online. The television series was $4.8$ gigabytes large and took $4$ hours to download.

**a** Find the size of the television series in gigabits. Use the formula $b = 8B$, where $b$ is the number of gigabits and $B$ is the number of gigabytes.

**b** Hence, find the average number of gigabits downloaded each hour.

**24** The life expectancy, $y$, of a person born in Japan is approximated by the equation $y = 0.27x + 72$, where $x$ is the number of years since 1970.

In what year will the life expectancy in Japan reach $95$ years?

**25** The amount of oil, $y$, imported to the United States from Canada in millions of barrels per day can be approximated by the equation $y = 0.105x + 1.34$, where $x$ is the number of years since 2000.

In what year will the approximate number of barrels imported from Canada be $5.5$ million per day?